

Sourim Banerjee

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Education

Purdue University

Ph.D. Student at School of Mechanical Engineering

August 2021 - Present

CGPA : 3.90/4

Indian Institute of Technology Kharagpur

B.Tech-M.Tech Dual Degree in Chemical Engineering

August 2016 - April 2021

CGPA : 9.21/10

Research Experiences

Purdue University

Advisor: Prof. Partha P. Mukherjee

August'21 - Present

Energy and Transport Science Lab(ETSL)

Doctoral thesis: Mesoscale Understanding of Instability at Solid-State Electrochemical Interfaces.

Indian Institute of Technology Kharagpur

Advisor: Prof. Monojit Chakraborty

August'20 - April'21

CoAdvisor: Prof. Partha Mukherjee, Purdue University

Chemical Engineering Dept, IITkgp

Masters' thesis : Mesoscale Origin of Voids in Metal Electrodes using kinetic Monte Carlo method.

Indian Institute of Technology Kharagpur

Advisor: Prof. Monojit Chakraborty

August'19 - April'20

CoAdvisor: Prof. Sunando Dasgupta

Chemical Engineering Dept, IITkgp

Undergraduate thesis: Metropolis Probability based kinetic Monte Carlo modelling of the protein nanoparticle drying pattern. (**Best B.Tech undergraduate thesis**)

Publications

1. **Sourim Banerjee**, Bairav S. Vishnugopi, Aditya Singla, Kautubh G. Naik and Partha P. Mukherjee.” Void Formation and Evolution Dynamics for Lithium Metal and Solid Electrolyte Interfaces.” ACS Applied Materials and Interfaces no. 18, (2026): 7981-1991.
2. **Sourim Banerjee**, Bairav S. Vishnugopi, and Partha P. Mukherjee. “Probing the Impact of Vacancy Diffusion on Void Dynamics at the Lithium Metal-Solid Electrolyte Interface.” Advanced Science (2026): e15827.
3. Min-Gi Jeong, Kelsey B. Hatzell, **Sourim Banerjee**, Bairav S. Vishnugopi, and Partha P. Mukherjee. “Temperature Impact on Lithium Metal Morphology in Lithium Reservoir-Free Solid-State Batteries.” PRX Energy 3, no. 2 (2024): 023003.
4. Debanjali Chatterjee, Aditya Singla, **Sourim Banerjee**, Anuththara Alujjage, Pretty Mitra and Partha P. Mukherjee. “MoChA: Modeling, Characterization & Analytics in Electrochemical Energy Systems.” ACS Energy Letters(2025): 3430-3436.
5. **Sourim Banerjee**, Bairav S. Vishnugopi, Guang Lin and Partha P. Mukherjee. “Mechanistic Insights into Instability from Single Void Dynamics at the Lithium Metal–Solid Electrolyte Interface.” (*Under Review*)
6. **Sourim Banerjee**, Ahmad Ahmad, Bairav S. Vishnugopi and Partha P. Mukherjee. “Role of Electro-chemo-mechanics in Dictating the Single Void Growth at Li-Solid-Electrolyte Interfaces.” (*In preparation*)
7. **Sourim Banerjee**, Ahmad Ahmad, Bairav S. Vishnugopi and Partha P. Mukherjee. “Plasticity and Creep-Mediated Chemo-Mechanics of Single Void Growth in Solid-State-Batteries” (*In preparation*)
8. **Sourim Banerjee**, Bairav S. Vishnugopi and Partha P. Mukherjee. “Investigating the Role of Secondary Metal Phase in Influencing Interfacial Stability in Li-Solid-Electrolyte Interfaces” (*In preparation*)

Conferences

1. **Sourim Banerjee**, and Partha P. Mukherjee. “Mechanistic Insights into Single-Void Growth and Morphology In Lithium Metal Solid-State Batteries.” In 2026 AIChE Annual Meeting in Minneapolis, MN.
2. **Sourim Banerjee**, Bairav Sabarish Vishnugopi, and Partha P. Mukherjee. “Influence of Surface Diffusion Interactions on Void Formation in Solid-State Batteries.” In 248th Electrochemical Society Meeting Abstracts, October, 2025.

3. **Sourim Banerjee**, Bairav Sabarish Vishnugopi, and Partha P. Mukherjee. “Role of Surface Diffusion Mechanism in Void Formation and Instability at Solid/Solid Electrochemical Interfaces.” In Materials Research Society Abstracts, Spring 2025.
4. **Sourim Banerjee**, Bairav Sabarish Vishnugopi, Kaustubh Girish Naik, and Partha P. Mukherjee. “Mechanism of Void Formation in Lithium Metal Solid-State Batteries.” In Electrochemical Society Meeting Abstracts 242, no. 4, pp. 383-383. The Electrochemical Society, Inc., 2022.

Computational Skills

Programming Languages

Matlab • Python • C++

Computational Methods and Solvers

Kinetic Monte Carlo • FEniCS (FEM) • FFT Micromechanics solver

Software Skills

GeoDict • OVITO • LAMMPS • L^AT_EX • Solid Works

• COMSOL Multiphysics

Courses

Core courses: Fluid mechanics, heat and mass transfer, reaction engineering, thermodynamics, transport phenomena, process modeling and simulation, instrumentation and process control, and microfluidics.

Relevant Courses: Fundamentals of Electrochemical Energy Systems, Materials Engineering Fundamentals, Introduction to Scientific Machine Learning, Calculus, Engineering Mathematics, Regression and Time Series model, Probability and Stochastic Process.

Teaching Experiences

Fundamentals of Electrochemical Energy Systems (ME 597), Prof. Partha P. Mukherjee Spring 2024

- Designed and upgraded the course’s lecture materials, homework, question papers, and instructed one independent lecture. Held office hours and graded papers for the class with a strength of 32.

Undergraduate Heat and Mass Transfer (ME 315), Prof. Thomas Beechem Fall 2024

- Held office hours, graded exam papers, and homework submissions.

Academic Achievements

- Selected for the prestigious **Purdue Trailblazers in Engineering** workshop aiming to provide first-hand training for faculty application process hosted at Purdue University in 2026.
- Awarded **Graduate Student Award for Outstanding Service Scholarship** in 2024 for aiding graduate student community development of Purdue through scientific community building.
- Awarded **Arrasmith Fellowship with Lozer Assistantship** in 2021 for extraordinary graduate school application at Purdue.
- **Department Rank 4** among 42 B.Tech-M.Tech Dual degree students in Chemical Engineering Department at IIT Kharagpur. Secured All India Rank **379** in prestigious **KVPY** (Kishore Vaigyanik Protsahan Yajana-translated to **Young Scientist Encourage Scheme**) scholarship and awarded prestigious **JBNSTS** (Jagadis Bose National Science Talent Search) scholarship, 2016.

Position of Responsibilities

- **Secretary & President, Electrochemical Society Student Chapter Purdue** August’21 - January ’25
 - Organized professional seminars, workshops, and poster symposium with enormous participation from the graduate students and faculty inside and outside Purdue University.
 - Under Sourim’s presidency (August ’24-January ’25), the chapter organized ‘Women in Sustainability’ and participated in the Fall 24 Research Expo organized by the Purdue Institute for a Sustainable Future.
- **Treasurer, Engineering Academic Career Club** August’23 - August’24
 - Managed organization funds, made budget and applied for a successful grant. Organized workshops and seminars related to future faculty application and post-PhD careers.
- **Mentorship, Student Welfare Group, IIT Kharagpur** August’18 - July’19
 - Mentoring 6 students of ChemE Department regarding academics as a part of Student Welfare Group(SWG).